ARAVINDHAN R

AI Engineer

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Professional Summary

AI Engineer specializing in Generative AI, with hands-on experience designing, building, and deploying scalable GenAI applications, Retrieval-Augmented Generation (RAG) pipelines, and Agentic AI systems capable of autonomous reasoning and task execution. Skilled across the full AI lifecycle, from LLM integration, prompt engineering, and model fine-tuning to multi-agent orchestration, using Python and modern AI frameworks. Backed by a strong foundation in deep learning and computer vision (CNNs) and a proven track record of shipping production-ready AI and full-stack products that improve automation, efficiency, and user experience.

Technical Skills

Languages: Python, JavaScript, C, HTML, CSS

GenAI Technologies: LLMs, Prompt Engineering, RAG, Multimodal RAG & GraphRAG, Knowledge-Augmented Generation (KAG), Knowledge Graphs, Semantic Modeling, Fine-tuning, Quantization, Re-ranking, Agentic AI, Multi-Agent Systems, A2A Protocols, MCP, Context Engineering, Guardrails, NLP

Frameworks & Libraries: FastAPI, LangChain, LangGraph, FastMCP, Flask, Next.js, Tailwind CSS, Bootstrap

Machine Learning & Deep Learning: Machine Learning, Deep Learning, CNNs, Graph Neural Networks (GNNs), Computer Vision, ML Algorithms

Models: GPT-4 / Codex, Gemini, Mistral

Data & Vector Stores: Pinecone, Neo4j, Firebase, REST APIs

AI Tools & Platforms: Hugging Face, Ragas, GitHub Copilot, Cursor, Claude Code, Jina AI, Selenium

Cloud & DevOps: AWS, Microsoft Entra ID (RBAC), Docker, Git

Other: Data Structures, Distributed AI Architectures, IoT

Work Experience

AI Engineer *Aug 2025 – Present*

Tata Consultancy Services (ID&A Unit) | Full-time *Chennai, Tamil Nadu, India · On-site*

- Designed and deployed a **RAG-based Test Case Generation microservice** using FastAPI and LangChain with a locally hosted, quantized Mistral model for data-security compliance and reduced latency/compute cost; integrated Pinecone for version-controlled vector embeddings of unstructured enterprise data, added a re-ranking mechanism for more accurate retrieval, and established continuous performance monitoring with Ragas, **reducing manual test-scenario creation effort by over 60%**.
- Developed an autonomous **Frontend Code Generation Agent** using LangGraph and a GPT-5.2-codex model in a multi-agent architecture that converts Figma designs into production-ready UI code; engineered a custom parsing algorithm to translate UI layouts into LLM-optimized structured data (mitigating context-window limits) and applied advanced context engineering and guardrails for security and coding-standard compliance, **cutting screen development lifecycles by 50–70%**.
- Built an **A2A (Agent-to-Agent)** communication layer enabling the agents to coordinate, reason collaboratively, and execute autonomous workflows, exposed through a remote **MCP Server** (FastMCP) that surfaces internal code-generation features as tools for AI assistants like Cursor and GitHub Copilot, feeding structured context to produce highly accurate, production-ready frontend code.
- Built a proof-of-concept **Knowledge-Augmented Generation (KAG)** system that automatically parses a relational SQL database to construct a knowledge graph, derives a semantic model over the resulting entities and relationships, and layers a graph-based RAG pipeline on top to deliver accurate, context-aware answers grounded in structured enterprise data.

- Building a proof-of-concept **train delay forecasting** system for an **Indian Railways (IRCTC)** client pilot: set up an event-driven data pipeline using SQL triggers on the client's operational database, where an AI agent picks up new records, converts them into graph-structured inputs, and runs them through a **Graph Neural Network (GNN)** inference engine to forecast train delays from live operational data.
- Serve as an **AI Friday Mentor**, guiding teams and individuals in rapidly prototyping AI-driven solutions to accelerate idea validation, product development, and technical implementation.

Founder-in-Residence *Jan 2024 – Sept 2024*
Zemuria (Kix News) *Remote · San Francisco, CA*

- Led product development of **Kix News**, an AI platform delivering unbiased news through comprehensive source collation and automated bias-checking.
- Built a news aggregation engine using Python, Gemini AI, Jina AI, and Selenium for automated data collection across diverse sources and RSS feeds.
- Designed data preprocessing pipelines to filter, validate, and categorize news content, significantly improving aggregation quality.
- Advanced the application through the investment phase to support enhanced feature deployment.

Projects

Mark Bot | *LLMs, Prompt Engineering, Full-Stack SaaS* · markbot.in

- Designed and built **Mark Bot**, an all-in-one operating system for content agencies that unifies production, operations, and billing into a single role-based dashboard for directors, managers, editors, and clients.
- Engineered an LLM-powered script-generation feature with Tanglish support, producing natural, performer-ready reel scripts and significantly cutting writers' scripting time.
- Implemented an 8-stage Kanban content pipeline (topic to publish) and automated, quota-aware invoicing that bills clients on actual delivery, reducing weekly operations meetings from 90 to 15 minutes.

Aura AI | *GPT-4 Fine-tuning, NLP, Generative AI, Machine Learning*

- Built an AI personal assistant focused on emotional support, personal growth, and productivity.
- Fine-tuned a GPT-4 model for response accuracy and personalized guidance, with empathetic chat responses grounded in therapeutic research.
- Implemented smart journaling and note-taking features that surface AI-driven insights to users.

Kavalar Eye | *CNN, Python, Flask, Docker, Hugging Face*

- Developed a deepfake-detection mobile application with a Flask backend.
- Trained a Convolutional Neural Network (CNN) model for accurate deepfake detection, collaborating with the team on model development.
- Dockerized the AI models with Flask APIs and deployed them on Hugging Face.

Education

Bachelor of Technology (B.Tech), Information Technology *2021 – 2025*
Sri Manakula Vinayagar Engineering College | CGPA: 7.29 *Puducherry, India*

Achievements & Awards

- **1st Place**: Hackverse Hackathon, conducted by PNB MetLife, India.
- **Certificate of Appreciation**: TCS AI Fridays, for valuable contribution as a Mentor (Chennai, Nov 2025).
- **Jury Member**: SDG AI Hackathon, organized by ETH Zürich and Nanyang Technological University (NTU), Singapore.
- **Internal Evaluator**: Smart India Hackathon (SIH) 2026, AI domain, evaluated 40+ prototypes and ideas.